TrES-3b

R-0012

Benchmark run · workflow W8-benchmark · record `2026-bench_TRES-3_260707` · created 2026-07-08T03:42:44+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461228.8682 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.056
Transit depth (Rp/Rs)^2	3.82 +/- 2.21 [%]
Orbital Inclination (inc)	83.46 +/- 2.94
Airmass coefficient 1 (a1)	5.17 +/- 3.13
Airmass coefficient 2 (a2)	-1.01 +/- 0.7
Scatter in the residuals of the lightcurve fit is	60.94 %
Best Comparison Star	None
Optimal Aperture	1.7
Optimal Annulus	6.0
Transit Duration (day)	0.061 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.056 vs 0.16309 (deviation 0.45σ)
Duration fit vs published	0.061 d vs 0.0567 d (deviation 0.14σ)
Mid-time O–C	0.0 min
Depth SNR	2.8
Residual scatter	60.94 %

Light curve

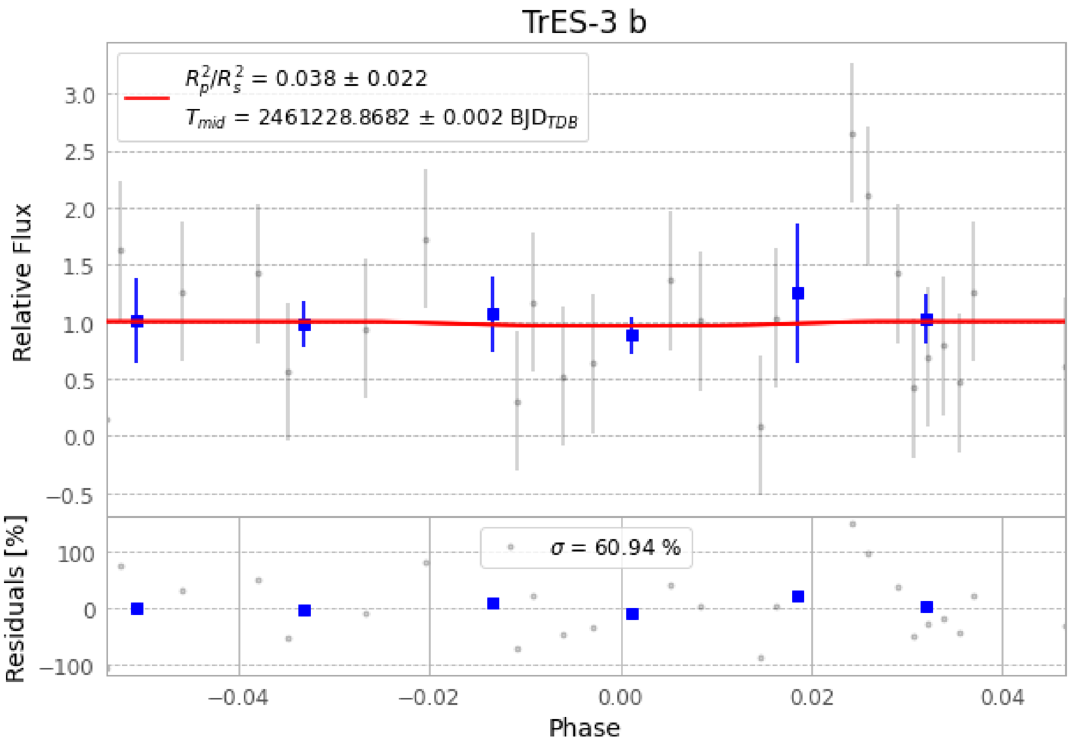

Figure 1 — FinalLightCurve_TrES-3 b_2026-07-07.png (SHA-256 41f3212cf8ee4440…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [296, 199], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 9991f209084c76fd…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

70 FITS frames (46 MB), `TRES-3260707070044.FITS` → `TRES-3260707102716.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-260707.log (01d7df7441bf…), FinalLightCurve_TrES-3 b_2026-07-07.png (41f3212cf8ee…), FinalLightCurve_TrES-3 b_2026-07-07.csv (43cae13c8be6…)

Compute resources

Wall time 139.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)